

# A Common Quotient-Subspace Obstruction to the Real HFP Conjecture

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## Status

This is a candidate partial result for Conjecture 5.2 of Aubrun and Muller-Hermes, *Limit formulas for norms of tensor power operators*, arXiv:2410.23063. It proves a common quotient-subspace obstruction to the Hilbert space factorization property (HFP). It does not solve the full conjecture.

## Source Question

Aubrun and Muller-Hermes define a pair  $(X, Y)$  of Banach spaces to have the HFP if every bounded operator  $\phi : X \rightarrow Y$  satisfies

$$\gamma_2(\phi) = \|\phi\|,$$

where  $\gamma_2$  is the factorization norm through Hilbert space. They conjecture in the real case:

A pair  $(X, Y)$  of real Banach spaces has the HFP if and only if  $X$  or  $Y$  is a Hilbert space.

The source records this as Conjecture 5.2 and gives an equivalent two-dimensional convex-body formulation as Conjecture 5.5.

as needed.

□

In the real case, we conjecture that there is no non-trivial pair with the HFP.

**Conjecture 5.2.** *A pair  $(X, Y)$  of real Banach spaces has the HFP if and only if  $X$  or  $Y$  is a Hilbert space.*

The pair  $(\ell_\infty^2(\mathbf{C}), \ell_1^2(\mathbf{C}))$ , when these spaces are seen as normed spaces over the reals, fails the HFP. This fact, which answers a question from [AMH24], can be seen using the following proposition.

**Proposition 5.3.** *If  $X$  and  $Y$  are non-strictly convex real Banach spaces, then the pair  $(X, Y)$  fails the HFP.*

It is a classical fact (see [Ami86]) that a normed space  $X$  is Euclidean if and only if all its 2-dimensional subspaces are Euclidean, and also if and only if all its 2-dimensional quotients are Euclidean. Using Proposition 5.4, this shows that in order to prove Conjecture 5.2 it suffices to consider the case  $\dim X = \dim Y = 2$ . This allows to formulate a concrete geometric question equivalent to Conjecture 5.2.

**Conjecture 5.5.** *Let  $K$  and  $L$  be centrally symmetric convex bodies in  $\mathbf{R}^2$ . Assume that for every linear map  $T$  such that  $T(K) \subset L$ , there is an ellipse  $\mathcal{E} \subset \mathbf{R}^2$  such that  $T(K) \subset \mathcal{E} \subset L$ . Then  $K$  or  $L$  is an ellipse.*

## Definitions

For a bounded operator  $u : E \rightarrow F$ ,  $\gamma_2(u)$  is

$$\inf\{\|a\| \|b\| : u = ba, a : E \rightarrow H, b : H \rightarrow F, H \text{ Hilbert}\}.$$

If no such factorization exists,  $\gamma_2(u) = \infty$ .

For an  $n$ -dimensional Banach space  $E$ , the Banach-Mazur distance from  $E$  to Euclidean space is

$$d(E, \ell_2^n) = \inf\{\|S\| \|S^{-1}\| : S : E \rightarrow \ell_2^n \text{ is an isomorphism}\}.$$

## Proof Intuition

The source proves that HFP descends to quotients of the domain and subspaces of the range. Therefore, if a pair  $(X, Y)$  had HFP and the same finite-dimensional non-Hilbert space  $E$  appeared both as a quotient of  $X$  and as a subspace of  $Y$ , then  $(E, E)$  would also have HFP. But the identity operator on  $E$  has Hilbert factorization norm exactly equal to the Banach-Mazur distance from  $E$  to Euclidean space. This distance is strictly larger than one when  $E$  is not Hilbertian, whereas  $\|\text{Id}_E\| = 1$ .

**Lemma 1.** *Let  $E$  be an  $n$ -dimensional real Banach space. Then*

$$\gamma_2(\text{Id}_E) = d(E, \ell_2^n).$$

*Consequently, if  $E$  is not linearly isometric to a Hilbert space, then  $\gamma_2(\text{Id}_E) > 1 = \|\text{Id}_E\|$ .*

*Proof.* If  $\text{Id}_E = ba$  factors through a Hilbert space  $H$ , then  $a : E \rightarrow H$  is injective and  $b|_{a(E)} = a^{-1}$ . Hence

$$\|a\| \|b\| \geq \|a\| \|a^{-1} : a(E) \rightarrow E\|.$$

Since  $a(E)$  is an  $n$ -dimensional Hilbert space, the right-hand side is at least  $d(E, \ell_2^n)$ . Taking the infimum over all Hilbert factorizations gives  $\gamma_2(\text{Id}_E) \geq d(E, \ell_2^n)$ .

Conversely, any isomorphism  $S : E \rightarrow \ell_2^n$  gives the factorization

$$\text{Id}_E = S^{-1}S$$

through the Hilbert space  $\ell_2^n$ , so  $\gamma_2(\text{Id}_E) \leq \|S\| \|S^{-1}\|$ . Taking the infimum over  $S$  gives the reverse inequality.

Finally, in finite dimension  $d(E, \ell_2^n) = 1$  holds exactly when the unit ball of  $E$  is an ellipsoid, i.e. when  $E$  is Hilbertian. Thus non-Hilbertian  $E$  has  $d(E, \ell_2^n) > 1$ .  $\square$

**Theorem 1.** *Let  $X$  and  $Y$  be real Banach spaces. Suppose that there is a finite-dimensional non-Hilbert real Banach space  $E$  such that  $E$  is linearly isometric to a quotient of  $X$  and linearly isometric to a subspace of  $Y$ . Then the pair  $(X, Y)$  fails the HFP.*

*Proof.* Assume for contradiction that  $(X, Y)$  has HFP. Proposition 5.4 of Aubrun–Muller-Hermes says that HFP is inherited by quotients of the domain and subspaces of the range: if  $(X, Y)$  has HFP, then  $(X/F, Z)$  has HFP for every quotient  $X/F$  of  $X$  and every subspace  $Z$  of  $Y$ .

By the hypothesis, choose  $F \subset X$  and  $Z \subset Y$  so that both  $X/F$  and  $Z$  are linearly isometric to  $E$ . Transporting the structure through these isometries, the pair  $(E, E)$  has HFP. Applying HFP to the identity operator on  $E$  gives

$$\gamma_2(\text{Id}_E) = \|\text{Id}_E\| = 1.$$

This contradicts Lemma 1, since  $E$  is not Hilbertian. Therefore  $(X, Y)$  cannot have HFP. □

**Corollary 1.** *If  $E$  is a finite-dimensional real Banach space that is not Hilbertian, then the diagonal pair  $(E, E)$  fails HFP.*

*Proof.* Apply Theorem 1 with  $X = Y = E$ . □

**Corollary 2** (Two-dimensional convex-body formulation). *Let  $K$  and  $L$  be centrally symmetric convex bodies in  $\mathbb{R}^2$ . If  $K$  and  $L$  are linearly equivalent and one of them is not an ellipse, then the hypothesis of Conjecture 5.5 in the source paper fails for this pair.*

*Proof.* Let  $T$  be a linear isomorphism with  $T(K) = L$ . If the hypothesis of Conjecture 5.5 held, there would be an ellipse  $\mathcal{E}$  such that

$$L = T(K) \subset \mathcal{E} \subset L.$$

Thus  $\mathcal{E} = L$ , which is impossible unless  $L$  is an ellipse. Since linear equivalence preserves being an ellipse, this contradicts the assumption that one of  $K, L$  is not an ellipse. □

## Verification Notes

No computation is used. The proof has two dependencies:

- the source paper’s inheritance proposition for HFP under quotients of the domain and subspaces of the range;
- the standard identification of  $\gamma_2(\text{Id}_E)$  with the Banach-Mazur distance from  $E$  to Euclidean space.

Both are exact norm statements, so no limiting or approximation step is hidden in the finite-dimensional obstruction.

## Limitations

The theorem only applies when the same finite-dimensional non-Hilbert space appears as a quotient of  $X$  and as a subspace of  $Y$ . Conjecture 5.2 remains open for arbitrary unrelated non-Hilbert pairs.

## Novelty And Search Bounds

Before packaging, the run’s cheap indexes were searched for arXiv:2410.23063, HFP, Hilbert space factorization, nuclear tensorization, tensor power operators, and the convex-body ellipse formulation. A local corpus search for the exact HFP/NTP terminology found the source paper and the earlier Aubrun–Muller-Hermes tensor-power paper. A bounded web search for the exact real HFP conjecture language did not reveal a later answer. The source paper already proves the non-strictly convex pair obstruction and the reduction to dimension two; this packet records only the common quotient-subspace obstruction above.

## References

- Guillaume Aubrun and Alexander Muller-Hermes, *Limit formulas for norms of tensor power operators*, arXiv:2410.23063.
- Joe Diestel, Hans Jarchow, and Andrew Tonge, *Absolutely Summing Operators*, Cambridge Studies in Advanced Mathematics 43, Cambridge University Press, 1995.

## Human Review Recommendation

Review as a small partial result. The main points to check are the application of the source inheritance proposition and the equality  $\gamma_2(\text{Id}_E) = d(E, \ell_2^n)$ .